Meeting Title: Visual Data Science Lab kickoff and assignment overview with Amal
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Transcript:
Amal Alnouri: and 26. My name is Amal Anouri. I'm a PhD student in the Visual Data Science Lab led by Professor Marc Steidt, and today is our first lab session. I'm going to walk you through the assignment. And the main goal of this session would be to give you an overview of what's needed from your side to do in this assignment, as well as show you the structure, how we structured the lab, the sessions, the kind of sessions we have, some information about the attendance, and so on. So. So let's get started. Um, I think so far you know that the course talks about the integration of visualization and artificial intelligence. So basically we're going to learn about how to use tools and techniques from each field. To help us perform tasks in the other fields. So from both sides, visualization for AI and also AI for visualization. So let's Start talking about the Vis4AI part. And maybe we start with a warm-up discussion. So I think that most of you, I'd say, has Have already like maybe trained an AI model or picked an AI model to perform a task? Can you raise your hand if you've ever done this? Of course, all of you. That's nice. So the other question, have you developed a strategy maybe to debug your model? Or maybe a strategy to convince someone else to trust it. Maybe you are dealing with stakeholders in a company and you want them to trust that your model is doing a good job or so. maybe you can tell us a little bit about your experience on this, either on debugging your model or convincing another one to use it. Yep, please. So for instance, I train models to perform on images from fields between crops and weeds, and my task is to train a model that can differentiate well between crops and weeds. So first I have to trust the model so that I can convince the customers to trust this model. Yeah, I usually will face some generalization issues in the fields because of the differences between fields, lighting conditions, cameras, and so on. Yeah, and how do you test this? How do you test if it's working well? Do you have any strategy? Yeah, I usually test on a separate project from different fields to measure the generalization rate for this model and to estimate If this model goes to a new field, will it still perform sufficiently well or not? Yeah, so do you do some manual annotations or do you use visualizations also to communicate this? The data is manually annotated first. The evaluation of the model is done on using some visuals, visualizations. Nice. Yeah. Maybe you also wanted to share something? Because they're learning the language, so we developed a little menu that talks to them in German and switch. Nice. And do you have any strategy to debug this or— We prefer like a hands-on approach with children. And we took basically the feedback of the teachers afterward. And do you use any visualizations maybe to communicate this feedback to a manager or? Do you try to create maybe some edge case? Edge cases to test with this model? No. Okay. Okay, nice. Thanks for sharing. So I think the other, the next question is something you all may relate to. Who finds it easy to understand machine learning concepts

through abstract mathematics? Nice. We already have 3 people who find it easy, but for the others who don't find it easy, there's good news. VIS can support you on this. So to address the first 2 questions, to debug your model and also to convince others that they can trust your model and use it, and also To help you to understand the complex concepts about machine learning, visualization can support you in those aspects. And the better news is that you're going to also use visualization in the assignment of this lab. To help you do such things. So basically, the main goal of our assignment is to give you hands-on experience using visualization to explain AI models and diagnose their failure modes. So first of all, there are some requirements. So first, you need to form groups of 4 members to work on this assignment. This is a must. And if you have not yet found your partners, you can use the discussion forum on Moodle. And please make sure to do this and find your group before the deadline. The deadline is on the 21st of October, so it's, I think, next Wednesday. If I didn't find a group till this deadline, will I be assigned randomly to some group? Uh, you should find a group. Why didn't you get it? Hidden, right? Match people together also that are not yet finished so that we get as many 4-people groups as possible. Yeah, but don't depend on us. Yeah, but you should try to open the big classes and only the problem is that sometimes the topics don't match me. Sometimes I text people and there are no free spots, so It can be that I don't find a group till this date. Yeah. But have you tried the forum on Moodle? Maybe you can, yeah, you can also post your topic if you're interested in something. People are doing this, yeah. So yeah, please try to find your group before the deadline. And then when you have a group, then we are asking you to answer 2 questions in this assignment. The first question is, how does the model behave? The second question is, what did the model learn? And the first question is phase 1 of this assignment. The other question is phase 2 of this assignment. And as you can see, the model is the hero of this story. So we need the model definitely. So the next or the second requirement is to have a model. And we are not going to force you to use a specific model, but whatever you are interested in, select model, select the task to train your model on, or a pre-trained model that performs a specific task. So basically, let me give you some guidelines on how to select this model and task. So you know that I think that the task is even more important than the model itself because One model can be trained to perform different tasks, but the task is the main thing here. So select the task that you are interested in and then select the model that can perform it. And here are some guidelines how to select these. These are not restrictions for you. So I'm going to mention some examples. These are not restrictions, but they are more like to steer you in this process. Yeah. So first of all, try to pick a well-defined task, for example, A classification task. It can be, you can classify images or even text, maybe sentiment analysis, maybe classifying cats and dogs, maybe medical images. You can, if you have specific knowledge in a specific domain, for example, medicine, biology, chemistry, you can pick a task from this field. And it would be also nice to see the diversity of the topics. Or maybe you can also, nowadays we have large language models or generative models in general. Large language models, for example, can perform broad, a broad range of

tasks. For example, you can focus then on problem solving, maybe on scientific question answering. Maybe you can also try to work on text summarization. There are endless tasks actually. And you are also, yep. Quite different in difficulty, like cat versus dog, very trivial versus like LLMs or— Yep. So some medical domains. Yeah. Is that in the grading to account for how difficult the task is that we chose? Or should we try to find something easy that we can do very well? Yeah. Yes, yes, yeah. So if you read through Moodle, then you can see at the end of the page we have a section for grading and we are mentioning this exactly. So we really value the creativity in this project. So definitely when you choose cat versus dogs, it's not like when choosing something more creative and complex, but you should also be able to analyze this case that you choose. So pick something that is reasonable for the time and for the effort you want to put in this project so you can deliver good results at the end. This doesn't mean also that those who pick cat versus dogs won't pass this assignment. They will pass if they do it in the right way, but I think for creativity there is a bonus also. I think you can make up for a simpler task with more creative solutions. Yeah. Because each task maybe saves you some time to set up the model and get data, then you can spend more time basically on the solution. Solution rather than a complex topic where it's hard to get like a— Yes. Probably, unless I guess you have some very specific domain that interests you, then I think the better choice might be to choose the more complex. Model and try to get something, right? So you need to find the correct balance. Mm-hmm. Sure, you can repeat the next question, I think. Oh yeah, yeah, sorry. Okay. Yes, and so After picking your task, you need also to pick a model that can perform it. So you are free also to decide whether you want to pick a discriminative model, and this also would depend on your task definitely, or a generative model. Even generative models can perform those traditional tasks like classification, for example, or so, so you are free to pick whatever you want. And next, of course, you're also free to pick a pre-trained model or maybe train your own model. Or maybe fine-tune a pre-trained model on a specific task. And also keep in mind that this model, most of you, I think they are going to run it on their own machines. So keep in mind the computational power needed to run this model. Don't pick something very large and then you don't know how to run it. And then Yes, so your analysis may also focus either deeply on one model. Or maybe compare several models. So maybe you answer those questions only for one model, or maybe you also answer this question, tell us how this model behaves differently from the other and what Did this model learn differently from the other one? So you do some comparison. It's up to you. And mind the time, of course. So Yeah, and definitely there is also a deadline for picking the task and model. It would be on the 28th of October. I think it's one week after the group formation deadline. And it's still not open in Moodle, so you cannot yet register your task, but we will share or announce the details later. Yes, so let's start maybe talking a little bit in more detail about Phase 1. So How does the model behave? How are we going to approach this question in our assignment? So basically, in this question, we are going to look at the model as a black box. There is some input that goes in, output comes out, and that's what we need. So we only— here we only want to look at the

inputs and output pairs. And do not consider what happens inside the model in this phase. So basically the main goal, or maybe the way to answer this question, is to design a visualization that helps you observe patterns in what goes in. And what comes out. I know this is so abstract, so let's maybe, yeah, let's maybe talk about this in more detail. So first of all, what should the visualization look like? So I'm asking you to design a visualization. What should this visualization look like? The first thing that might come into your mind is maybe to calculate some performance metric and maybe then report this performance metric in a specific, maybe bar chart, confusion matrix, whatever you want. But this is wrong. This is not what we are asking you for. What we are asking you for is more like using visualization as an investigative tool to explore patterns in model behavior. And this is still so abstract, so I'm going to show you an example. And it's a very generic example. It's not something to do again in your project, but it's just an example. So don't overfit to it. I think I'm not sharing now. So Okay. Yeah, so basically in this interactive visualizations, the authors are like addressing BERT models. A kind of transformer models. And, you know, these models, when you give them, just like any LLM, when you give them a sentence and then a blank, they try to complete this sentence. And basically, these models, of course, are trained on like a very big amount of text. So they have some vocabularies. For example, the authors mentioned that this model that they are studying here learned 30,000 words. Words and also, so the case that we are looking at here, they start a sentence, for example, in Texas, they like to buy and then a blank. And then another sentence, in New York, they like to buy, and then a blank, and the model should continue this blank. And the authors aim from this experiment to see whether the model associates specific items with Texas and other items with New York. So basically, when the model wants to continue this, what they do actually is that they assign a probability for each word they have. Some of these words would be assigned to higher probabilities than others, and this is what they want to measure. So if some word, for example, like books is assigned higher with higher probability in Texas, this means maybe the model thinks that people in Texas buy books more than people in New York. So to show this, they have a very simple visualization, but it's also an effective one. It's only a scatterplot. Each point in this scatterplot is a word. An item. That the model can say, can mention. And we have Texas, or we have New York in orange, and we have Texas in green. And then we have the likelihood to mention this word after, in this blank after New York is the x-axis and the likelihood to mention this word in this blank after Texas is the y-axis. And then we can see that the words here. are more associated with New York while the words here are more associated with Texas. So we can see that the model thinks that people in Texas would buy beer. Land, guns. But people in New York would buy books. Um, Clothing, art. And you can also play around with this. For example, let's try Austria. Let's see Austria and New York. Yeah. So in Austria, they buy wine, horses, flowers, perfume. That's what the model thinks. You can also increase the number of tokens shown in this visualization. So it's, it's really very simple, but it's also very nice and effective. Right. Yeah. So yeah, you can understand stereotypes by this. Yeah. So let us go back to the slides again. About. Above this one. I

think there is nothing speaks against it. Yeah, if you have a way to run the model. Yeah. Okay. I think we should share again. So let me first Yep. Yes. Um. Yeah, so we are expecting something more like this, not like that. And of course, more examples in the upcoming sessions. Yes, and then this is about the visualization. Then we are asking you to use this visualization. To help you observe patterns in what goes in and what comes out from the model. So what kind of patterns are we expecting you to find? These are also examples. So let's start maybe the pattern that would come into your mind. Would be errors. Of course, errors are patterns. So for example, this is a very common example. If you are— so when you are maybe training a classifier to classify images of wolves and huskies, and then most of your wolves are with like maybe snow backgrounds, and then maybe the model would learn to characterize wolves from, or yeah, from the background, from the snow instead of real characteristics so for example, such an error, it's also a behavior that you can detect. And report in your assignment. Another thing, it's to look at not only the error, whether the prediction has been flipped, but also the confidence change. So for example, if some of those images are like, they are detected as huskies, for example, but with 0.52 confidence. So they are still classified correctly, but the confidence is too low right. This is also a very interesting pattern to look at. Other kinds of patterns? Biases. I think you are familiar with biases. For example, a translation model tends to translate gender-neutral occupation into male for engineers and female for teachers. The same thing, for example, for A language model, if you tell a language model, I am a nurse, and then ask them a question, maybe they start address you with female pronouns. This is also a kind of bias. And also for text-to-image models, when you tell them to generate a doctor, maybe they generate males. Another kind of behavioral pattern is hallucination. It's also, I think you are also familiar with it. It's specifically for LLMs. It's when you ask an LLM a question and confidently it answers wrongly. So maybe again, like for this example, Maybe you find that your language model is hallucinating more in specific scenarios than others, or maybe you find that there is a specific model that is hallucinating more than the other model. This is also a kind of behavior to report. There is also safety violation. So it's, so you know that, for example, also again, language models, it has some safety guidelines or policies for example, when you ask them directly, About a restricted request, they might not answer. But when you tell them in a fictional story, a character asks for the same restricted request, maybe they would tell you, here's the response. And also, this is a very interesting kind of pattern to report. Another thing also is shift in output characteristics. So my examples are here focused on text, but they can also work for images. So here we have 2 prompt variations. You ask the LLM to summarize this article and then they give you an answer. But when you ask them politely, maybe they switch their tone or maybe they give you more detailed answer. These are also characteristics. Shift in characteristics, and they can also count as change in the model behavior. They are interesting also. To study. Okay, and these are again only examples, so please be creative. There can be endless number of behaviors that you can find. Yeah, so as you see, for detecting those behaviors, you need a dataset of inputs. You feed the model with this dataset and then you observe how the

output would be. So what kind of input data should you use? You can definitely use some official test data released with the model. Usually we have such datasets, a test dataset. You can also use an established benchmark for a task. So for example, I think there are too many benchmarks, especially for language models, let's say. So benchmarks to measure the model's performance maybe on logical tasks. Yeah, maybe on text summarization. You can find such datasets. And more interesting thing is to use maybe a synthetic dataset. Either you design it, either you generate it. Or maybe you find it, it's already there designed or generated for you. And then you investigate particular hypotheses with it. So like what, for example, maybe you can, maybe your dataset Your training dataset, you found out that they do not have such edge cases. They are all about dogs in maybe in nature. But you don't have such an edge case, so you generate them. Maybe you do some perturbations, then you don't even need to generate those images, but you can do this by code. Maybe you add blur, maybe you add grayscale images, maybe you also do some counterfactual. examples. So you switch something you think the model is depending on, like the background here. And for counterfactuals, they are usually examples from the same distribution, but for perturbations, they are outside the distribution of the training data. Um, And then you can also An interesting— another interesting thing is to use the training data itself as a source of your hypothesis. So for example, I think we mentioned this before. If you figured out that there is maybe an underrepresented class in this dataset or an underrepresented case like this edge case in the dataset, then you can hypothesize something and create edge cases or and observe how the model performs on these cases. So if you want something really interesting, some interesting findings, those 2. Yeah, would really generate interesting findings for you. So for now, we have talked about phase 1. So let's talk a little bit about phase 2. In phase 2, you are supposed to answer the question, what did the model learn? So in the first phase, we decided to keep this model or this black box closed, but in phase 2 now, we open this black box and we are interested in learning what did the model learn. So you need to explore the internal representation of this model, and what is expected from your side is to design a visualization that helps you observe patterns in how the model produces its outputs. And basically, I think you're going to be introduced to these kind of techniques that helps you know what's going on inside the model later in the lecture. So Again, I'm asking here, what should the visualization look like? So there's— this is the wrong way to do it, and this is called saliency maps. And what we see here, we have here maybe an image classifier. And the spots that we see here are where this classifier looks at when they predict. That this is maybe a bird or a specific kind of bird. So these are the pixels that activate its prediction. Yeah, so what we don't want you to do is to pick like, maybe you have cherry-picked examples and show them and tell us, okay, it seems like the model is focusing on those areas when doing the prediction. Okay, nice, but you're not going to pass the assignment like this. What we want you to do is more like, again, to ask a question, analyze, and use the visualization to answer this question. So maybe the ideal way to do it is to pick a behavior from phase 1 and try to explain this behavior using what the model learned in phase 2. So for

example, in phase 1, The model makes more mistakes on certain backgrounds, and then in phase 2, you tell us, does the model rely on the background features when producing these predictions? So in this case, if it really relies on them, we need— we would say the saliency map on the background instead of the same of the object itself. Um, yeah. So again, more examples in the upcoming sessions, and after you are familiar with these techniques more in the lecture, so I'm going to keep this for later. And now we are done with the Vis4AI part of the assignment. Do you have any questions? Any questions on Zoom? No. I think you've already answered. Yeah, it's for— sorry. Yeah. Okay. Yeah, so let's talk a little bit about the AI for visualization part. And again, Was there a question? Yeah, sure. To ask, maybe this will be covered later. Because depending on the model, it might not be possible to upload the model individually, right? Should we then also disclose where we downloaded the model, or— Yeah. Not in this presentation. What you need to submit, we will cover later during the course, but I don't think you need to upload your model, I guess. Yeah, maybe if you have a link to the model, you can include it. Basically, I think ideally we would have a GitHub repository. And so you can also add the link to it instead of uploading the code. We will talk about this also later. Yeah. So let's again start with the discussion. I think the answer would be yes by all of you. Do you collaborate with LLMs to perform tasks? Definitely you do. And then the other question. Maybe it's also yes. Have you ever created charts to explore or communicate your data? I guess most of you have experience with the charts. Then comes the third question, have you tried to use generative AI to assist you in designing visualization? Have you ever done this? Can you tell us about your experience? How did you use it? Yeah. Analyzing like basic error statistics or something during the training process of an actual AI model. These are like often quite basic visualizations and it's Like fast to just ask an LLM to just give it the data, see how it's saved, and just visualize it, not even giving any instructions because these are such basic visualizations that it works quite well. And it would take a lot more time to first think about how it should look or maybe even look up the documentation on which command actually isn't that perfect. Nice. Yeah, like easy task. I think you could just say this is my data and it does reasonably. Yeah, nice. Thanks. Do you want to add something? I can definitely say for sure that the error object is being formed when we used to try to do that. During my early studies, it was actually very hard to create proper, good-looking plots. Maybe well-trained on that, and one could actually create the plot based on some data, but it would look not nice, at least. First, you would definitely have to analyze the planning, but with modern models, it's quite easy to do this. Some specific and simple plots. Complex. That's a nice observation. Thanks for sharing. Do you also want to— want to add something? Start with writing the prompt, seeing the result, realizing I don't like generative AI, and ask it to write a code so I can do the thing. Yeah, these are nice strategies actually. And yes. I have also a list of similar things here. Yes, maybe you can use AI to brainstorm and sketch designs for visualizations. Maybe you can provide them with the data and generate a visualization if this works well. Maybe you can also provide or describe your data or design to the model and generate a visualization code instead of an image. Maybe you can also provide

the LLM with your Visualization and ask them to explain or annotate this visualization. Maybe you have a cluster of data items and you want to know how these data items are different from the data items in another cluster, and then maybe you can use the LLM to describe this difference or so. Another thing you can also do is to transform your data to be suitable for visualization. For example, you have unstructured text and you want to extract things from there. Maybe you use an LLM to do this and too many other use cases. So visualization or AI can also support visualization in these aspects. Um, Yeah, so basically in this part of the assignment, You're going to explore how generative AI can support and influence also the visualization design and analysis process. So we would like you to use generative AI. To help you design your visualizations and maybe even interpret them. And then at the end, you need to reflect back on your experience and report the lessons you learned from this. And tell us how the generative AI influenced your process, your design and analysis process. And basically this is going to work first by you form your groups of 4 members and then you're going to form 2 subgroups inside the same group. A pair of you would work aided by AI and the other pair would work without AI. You're gonna work on the same Of course, you're going to align together. So work on the same task, on the same model, do some high-level alignment. Don't force both of the pairs to do the exact same analysis and also don't make it like a race because it's not like this. So try to align at the high level. And then this is for phase 1. For phase 2, you will switch roles. So yeah, you will switch roles. So those pairs who did not work with generative AI before would work with it and the pairs who worked with it won't work with it for phase 2. Sorry, this is phase 2, but it's phase 1 in the slides. Yeah, and finally, of course, All of you need to discuss with each other and then come up with the lessons learned from this experience and collect all of your findings and put them in the same presentation. For us. groups that would use AI. How much of AI should we use, how far should we go? Because we also discussed that in lecture. Yeah, partially discussed it in lecture, and the professor just like an hour ago told us that there is a new tool with the Claude. Well, basically, I'm afraid that it might be able to do the whole work for us, which is fine. Yeah, yeah, yeah, definitely there is such a Yeah. Of course, you are free to do whatever you want, maybe generate everything in one prompt, but at the end, I think the challenge is to come up with those. So the main thing is not to, like, it's not just to use AI, but also to reflect on this usage. So you should tell us what prompts you used. You should also tell us what you should evaluate it, whether, like, did you accept it? Why did you accept it? Did you reject it? Did you refine the code maybe, or Yeah, you should reflect on this, and also we are expecting more from you as the generative AI pair. For example, we are expecting you to do some interactive visualizations that we are maybe not expecting from the other pair, maybe to broaden the analysis also. Yeah. So you should consider. This. Um. Um, I have a question regarding the group that doesn't use AI. Like, would they be still allowed to use AI to ask like technical stuff? Like, let's say, how do I create a topic? Yeah, yeah, of course. Like for maybe light code completion or yeah. Completion, yeah, and then maybe also searching documentations or so, it's okay to use it, yeah, definitely. You can't,

for example, generate code. You can't use it to understand your visualization. Don't use it to brainstorm also visualization designs also. Don't use it to generate visualizations themselves, like an image. Yeah. No. No. But I think the limit is rather low, right? So you should write as little as possible at this point, try to see how far you can get without it. And we don't care too much, maybe that's something you're gonna add at some point. If you cannot get very far and you realize you're stuck with some simple chart that you could find in the docs for some study plotting, okay, that's it, right? That's not bad for the assignment necessarily. As long as you reflect on that and compare with the other solution, so you should not Like feel forced to try or I don't know, feel tempted to use AI to get a better result. Point is really to see how far you can get with coding, even if that is not very far. Yeah, considering that both of you worked in the same time, I mean, took the same time to work on this project and also put the same effort in it. Yeah. So don't also mean to just generate or just come up with static charts or so. Put your effort in making those charts and Okay, yep. Question about the report, because like if there are like people in this files and we are creating In-depth interactive visualizations, like step by step, there will be a lot of prompts and stuff. How long should this report be? So like minimum 10 pages, maximum 20, or just one? Or like, what is the dimension of this report? Yeah, it's not a report, it's a record of those. So maybe you can put it in an Excel sheet or whatever, you know. No, no, these are the things we need. You should report these things. Yeah. So you should record these, the prompts and so, and you should report these in your maybe a presentation. Probably not. Um, Yeah, so in the final— yep, please. It's mentioned previously that each frame is one thumbprint. Mm-hmm. Awesome. Combine, merge with ours. How to be I think it's your job to sit together and discuss what each of you found and also do a comparison. Tell us, this is what come up from AI-assisted design and this is What's come up from without AI, for example, it's up to you to come up with the correct narrative. Go is not necessary. Create, join, common stuff that combines the ideas from both. But to understand how groups approach the problem differently and if they were successful in different ways or not, it's more reflection of how things went than presenting your finds. Yes, the 2 groups can do their own thing, I guess. Yeah, but you should also align at the high level, at least, you know. So you should know what's going on in the other pair. For example, if they decided to compare 2 models or they decided just to pick one model and analyze it in depth, you should do the same. Not that one of you go with one model and the other group picked 4, 5 models, you know. Like, with aligning on a high level, does this mean that we should make sure that We are doing different stuff because for some tasks, like for example language modeling, they're quite similar plots that are like the, I don't know, the low-hanging fruits you would do. Like, I mean, not low-hanging fruit in the sense of easy, but like something like you showed, like some embedding space or anything. This is quite what, like, almost which would come to mind probably. Should we align so far that we say, okay, we are doing this and you're doing something different? No, don't force it. Model and data. No, no, don't force it, especially those design decisions, for example. Yeah, it's okay if they went with another, if they go with another design decision than you, it's okay. This is not a

high-level— no, it's not also— no, it's not necessary. No. No, no. It's not necessary as well. Versions, right? The one version you would see, okay, maybe the one group just creates the spare embedding because that's how far they get, and the other group builds a whole interactive white-coded prototype around it. There is a code that does not work. No, because then you can maybe reflect on it more easily. I don't know. But if the idea, like the ideation is already part of how you use AI, then maybe the AI would bring you to a different track and that's your insight, right? That the AI could end it up in something completely different. Yes, exactly. Yeah. Okay. So that's all about the AI and Vis. Is there still any questions in your mind? Should I also check the Zoom? Should I check this one? Are there too many questions? Okay, thank you. Yeah. So let me show you again the course timeline. So you had the kickoff a week ago. And then we have the group formation deadline on the 21st of October. A week after this, we have the model selection deadline. And then we have the midpoint deadline, and this is the deadline for phase 1. So here phase 1 should end and you should report your results, submit them, and then we have the final deadline about a month after it. Directly after the final deadline, we have the presentations. We have— so before this, during this period, we have sessions in the lab where I'm going to also share some examples with you, and there is also we will have feedback sessions. We are going to share the details about these feedback sessions later, so you are going to be able to book a slot, maybe a 10-minute slot where you bring your results and Maybe you discuss it with us, you give— we give you some feedback on this. The attendance here is not mandatory. But for the presentations, it's mandatory. So we have 5 slots for the presentation. You should at least attend one of these slots, and in these slots Of course, we have too many groups. We cannot let you all present, but we are doing a big random set of these groups, and those groups would present live. Yeah, but for the other groups, we would grade based on your recorded presentation. Um. So no deadline extensions because our timeline is tight, so we need to finish before Christmas. So do not assume that we are going to extend those deadlines. Keep in mind that they are strict deadlines. Also, We expect you to— so it's the attendance is not mandatory, but we expect you to keep up with recorded labs and also lectures to complete the assignment successfully. Without them, I don't expect you to, to be able to complete the assignment successfully. Yep. With your presentations, we go. Both online and in person. It would be nice if you are here in your slot, so you will register also for the slot that you prefer, and then if you, if you are randomly selected, you need to present, it would be also nice to have you in person here in that slot. Yeah, but you can also present online. Yeah. So Do you have any other questions to address in this session? If no other questions, then We are done. Them: Yeah, thank you, but I've just realized it's not started. And I think you can